

Practical week-3

TASK-1

Develop a C program using fork() that creates a parent and child process. Display the Process ID(PID),Parent Process ID(PPID),and process states at different stages of execution

```
gopinath@GOPINATH:
GNU nano 8.7.1 task1.c

#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
pid_t pid;
printf("Parent process started\n");
printf("Parent PID :%d\n",getpid());
printf("Parent PPID :%d\n",getppid());

pid=fork();

if(pid<0){
perror("fork failed");
return 1;
}
else if(pid==0){
printf("\n--- Child Process ---\n");
printf("Child PID : %d\n", getpid());
printf("Child PPID : %d\n", getppid());
printf("Child State: Running\n");

sleep(5);

printf("Child State: Terminating\n");
exit(0);
}

else
{
// Parent process

```

[Read 46 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

```

else
{
    // Parent process
    printf("\n--- Parent Process ---\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    printf("Parent State: Running\n");

    printf("Parent State: Waiting for child...\n");
    wait(NULL);
    printf("Parent State: Child terminated\n");
    printf("Parent State: Terminating\n");
}

return 0;
}

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

gopinath@GOPINATH:

```

gopinath@GOPINATH:~$ nano task1.c
gopinath@GOPINATH:~$ gcc task1.c -o task1
gopinath@GOPINATH:~$ ./task1
Parent process started
Parent PID :656
Parent PPID :356

--- Parent Process ---
Parent PID   : 656
Parent PPID  : 356
Parent State: Running
Parent State: Waiting for child...

--- Child Process ---
Child PID    : 657
Child PPID   : 656
Child State: Running
Child State: Terminating
Parent State: Child terminated
Parent State: Terminating

```

TASK-2:

To observe the different process states-Ready, Running, Waiting, and Terminated-using Linux monitoring tools such as ps, top , and /proc.

Sol:

gopinath@GOPINATH:

GNU nano 8.7.1

process_states.c

```

#include <stdio.h>
#include <unistd.h>
int main() {
    printf("Process started!\n");
    printf("PID: %d\n",getpid());
    printf("Running state...\n");
    sleep(2);

    printf("Waiting/Sleeping state...\n");
    sleep(10);

    printf("Process terminating...\n");

    return 0;
}

```

[Read 15 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

```
gopinath@GOPINATH:~$ gcc process_states.c -o process_states
gopinath@GOPINATH:~$ ./process_states
Process started!
PID: 832
Running state...
Waiting/Sleeping state...
Process terminating...
```

Sol:

gopinath@GOPINATH:

```
gopinath@GOPINATH:~$ ps -o pid,ppid,state,cmd -p 832
  PID   PPID  S CMD
gopinath@GOPINATH:~$ top
top - 13:56:56 up 51 min,  1 user,  load average: 0.00, 0.02, 0
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,
MiB Mem :  3758.6 total,  3113.5 free,   462.1 used,   258.
MiB Swap:  1024.0 total,  1024.0 free,    0.0 used.  3296.
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
1	gopinath	20	0	24164	15472	11632	S	0.0	0.4
2	gopinath	20	0	3180	2200	2072	S	0.0	0.1
7	gopinath	20	0	3180	2048	1940	S	0.0	0.1
56	gopinath	19	-1	50428	16784	15484	S	0.0	0.4
88	systemd+	20	0	22416	14460	11968	S	0.0	0.4
103	gopinath	20	0	35312	12228	9188	S	0.0	0.3
120	gopinath	20	0	2888	1948	1820	S	0.0	0.1
121	gopinath	20	0	4472	3132	2884	S	0.0	0.1
123	message+	20	0	8820	5344	4676	S	0.0	0.1
155	gopinath	20	0	44096	29764	14624	S	0.0	0.8
162	gopinath	20	0	18424	9440	8204	S	0.0	0.2
177	syslog	20	0	220548	5932	4648	S	0.0	0.2
220	_chrony	20	0	23892	11340	7136	S	0.0	0.3
236	_chrony	20	0	12096	2460	1812	S	0.0	0.1
267	gopinath	20	0	5492	2768	2584	S	0.0	0.1
269	gopinath	20	0	123460	32676	18060	S	0.0	0.8
353	gopinath	20	0	3184	1104	980	S	0.0	0.0
354	gopinath	20	0	3200	1244	1108	S	0.0	0.0
356	gopinath	20	0	6268	5528	3856	S	0.0	0.1
357	gopinath	20	0	8644	5536	4696	S	0.0	0.1
406	gopinath	20	0	22340	13272	10992	S	0.0	0.3
408	gopinath	20	0	22916	4076	2076	S	0.0	0.1
440	gopinath	20	0	6136	5416	3864	S	0.0	0.1
902	gopinath	20	0	8164	5996	3896	R	0.0	0.2

```
gopinath@GOPINATH:~$ ./process_states &
[3] 1261
Process started!
PID: 1261
Running state...
gopinath@GOPINATH:~$ Process terminating...
Waiting/Sleeping state...
ps -o pid,ppid,state,cmd -pps -o pid,ppid,state,cmd -p 1261
  PID   PPID  S CMD
1261    356  S ./process_states
[2]-  Done                  ./process_states
```